

# Mathematical and Computational Methods

## Exercise Sheet 2: Complex numbers I: algebra and geometry

**Tutorial route** (►, in this order, about 80 minutes): 1.3, 2.1(a,c), 2.3, 2.4, 2.5(a–c), 2.7, 2.10. Physics groups may replace 2.4(b) with 2.6. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★ ★ challenge. Give exact answers unless a decimal is asked for.

## 1 Warm-up drills

**Exercise 1.1** (★ drill — powers of  $i$ ). Simplify the powers of the imaginary unit

$$i^3, \quad i^4, \quad i^{15}, \quad i^{2026}, \quad i^{-1}, \quad i^{-7}.$$

**Exercise 1.2** (★ drill — Cartesian arithmetic). With  $z = 3 + 2i$  and  $w = 1 - 5i$ , find  $z + w$ ,  $z - w$ ,  $zw$  and  $z^2$ .

**Exercise 1.3** (► ★ drill — parts, conjugate and modulus). For  $z = 12 - 5i$  find: (a)  $\operatorname{Re} z$  and  $\operatorname{Im} z$ ; (b)  $\bar{z}$ ; (c)  $|z|$  and  $z\bar{z}$ ; (d)  $\operatorname{Im}(iz)$ .

**Exercise 1.4** (★ drill — division). Write  $\frac{4 + 3i}{1 + 2i}$  in Cartesian form  $a + bi$ .

**Exercise 1.5** (★ drill — quadratics over  $\mathbb{C}$ ). Solve  $x^2 + 1 = 0$  and  $x^2 + 25 = 0$  over  $\mathbb{C}$ .

**Exercise 1.6** (★ drill — an Argand sketch). On a single Argand diagram, plot the points  $z_1 = 3 + i$ ,  $z_2 = -2 + 2i$  and  $z_3 = -1 - 3i$ , together with the sum  $z_1 + z_2$ . Label each point.

**Exercise 1.7** (★ drill — polar and back). Convert to polar form  $r(\cos \theta + i \sin \theta)$  with principal argument: (a)  $1 + i$ ; (b)  $2i$ ; (c)  $-3$ . (d) Convert  $z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$  to Cartesian form.

**Exercise 1.8** (★ drill — the polar rules). Let  $z = 3(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$  and  $w = 2(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12})$ . Using the polar rules, find  $zw$  and  $z/w$  in Cartesian form.

## 2 Tutorial problems

**Exercise 2.1** (► ★★ core — the conjugate and modulus rules). Let  $z = 2 + 3i$  and  $w = 1 - 4i$  (as in the notes).

- Compute  $zw$  and  $z/w$  in Cartesian form.
- Verify the conjugate rule  $\overline{zw} = \bar{z}\bar{w}$ .
- Verify the modulus rule  $|zw| = |z| |w|$ .

**Exercise 2.2** (★★ core — two general proofs (maths)). (Maths stream — general proofs.) Let  $z = a + bi$  and  $w = c + di$  with  $a, b, c, d \in \mathbb{R}$ .

- Prove the conjugate-of-a-product rule  $\overline{zw} = \bar{z}\bar{w}$  directly from the definition  $\overline{a + bi} = a - bi$ .

(b) Using (a) and  $|u|^2 = u\bar{u}$ , prove that  $|zw| = |z| |w|$ .

**Exercise 2.3** (► ★★core — *conjugate pairs*). Solve over  $\mathbb{C}$ , and confirm that the roots form a conjugate pair: (a)  $x^2 + 2x + 10 = 0$ , by completing the square; (b)  $2x^2 + x + 1 = 0$ , by the quadratic formula.

**Exercise 2.4** (► ★★core — *plotting and distance (data science)*). (a) Let  $z = 3 + 2i$  and  $w = -1 + 2i$ . Plot  $z$ ,  $w$ ,  $\bar{z}$ ,  $z + w$ ,  $z - w$  on an Argand diagram, and compute the distance  $|z - w|$  between the points  $z$  and  $w$ .

(b) (*Data science — distance between records.*) Two data points in the plane are encoded as  $p = 2 + 5i$  and  $q = 6 + 2i$ . Compute the Euclidean distance  $|p - q|$  between them and the midpoint  $\frac{1}{2}(p + q)$ .

**Exercise 2.5** (► ★★core — *principal argument*). Convert each number to polar form, giving the principal argument  $\text{Arg } z \in (-\pi, \pi]$ . Sketch each to fix the quadrant.

(a)  $1 + i$ ; (b)  $-1 + i$ ; (c)  $-2 - 2\sqrt{3}i$ ; (d)  $\sqrt{3} - i$ ; (e)  $-5$ .

**Exercise 2.6** (★★core — *AC impedance (physics)*). (*Physics — AC circuits.*) A series circuit has complex impedance  $Z = 4 + 3i$  (in ohms). It is driven by a voltage  $V = 10$  (volts, taken as a real reference).

(a) Find  $|Z|$  and the phase angle  $\text{Arg } Z$ .

(b) Compute the current  $I = V/Z$  in Cartesian form and its magnitude  $|I|$ .

**Exercise 2.7** (► ★★core — *polar product and quotient*). Let  $z = \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$  and  $w = 2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ . Find (a)  $zw$  and (b)  $z/w$ , giving each in polar form with the principal argument and in Cartesian form.

**Exercise 2.8** (★★core — *multiplication by i (with phasors)*). (a) For  $z = 2 + i$ , compute  $iz$  and confirm geometrically that multiplying by  $i$  is a quarter-turn anticlockwise that preserves length.

(b) Which single complex number rotates a point by  $\frac{\pi}{2}$  anticlockwise and doubles its distance from the origin?

(c) Apply a clockwise quarter-turn to  $3 + i$  (i.e. multiply by the number of modulus 1 and argument  $-\frac{\pi}{2}$ ).

(d) (*Physics — phasors.*) A voltage phasor  $V_1 = 6 + 8i$  is joined by a second phasor  $V_2$  of equal amplitude turned a quarter-turn anticlockwise, i.e.  $V_2 = iV_1$ . Find  $V_2$ , the resultant  $V = V_1 + V_2$  in Cartesian form, and its amplitude  $|V|$ .

**Exercise 2.9** (★★core — *true or false*). (*Conceptual — true or false, with a one-line reason.*)

(a) For every  $z \in \mathbb{C}$ ,  $z^2 \geq 0$ .

(b)  $|z| = |\bar{z}|$  for all  $z$ .

(c) If  $z + \bar{z} = 0$  then  $z$  is purely imaginary or zero.

(d)  $\text{Re}(zw) = \text{Re}(z) \text{Re}(w)$  for all  $z, w$ .

(e) There is an ordering  $<$  on  $\mathbb{C}$  compatible with its arithmetic — that is, one respecting  $+$  and  $\times$  (if  $x > 0$  and  $y > 0$  then  $x + y > 0$  and  $xy > 0$ ), so that every nonzero square is positive.

**Exercise 2.10** (► ★★challenge — *an equilateral triangle*). (*Rotation/geometry.*) Two vertices of an equilateral triangle lie at 0 and 2 on the real axis. Use multiplication by a complex number of modulus 1 to find the two possible positions of the third vertex.

### 3 Further practice (home)

**Exercise 3.1** (★★core — solve for real  $x, y$ ). Find the real numbers  $x, y$  satisfying  $(x+yi)(3-i) = 7+i$ .

**Exercise 3.2** (★★core — division). Simplify to Cartesian form: (a)  $\frac{1+2i}{3-4i}$ ; (b)  $\frac{i}{1+i}$ .

**Exercise 3.3** (★★core — a real cubic from one root). Given that  $1+2i$  is a root of  $x^3 - 4x^2 + 9x - 10 = 0$  (real coefficients), find the other two roots.

**Exercise 3.4** (★★core — loci and a region in the plane). Describe geometrically, and sketch, the set of points  $z \in \mathbb{C}$  satisfying

$$(a) |z| = 2; \quad (b) |z - 1| = 3; \quad (c) |z - i| = |z + i|; \quad (d) |z - 1| < 2.$$

For (d), shade the region and show clearly whether its boundary is included.

**Exercise 3.5** (★★core — one product, two ways). Compute the product  $(1+i)(\sqrt{3}+i)$  in two ways — by Cartesian expansion and by the polar rule — and hence write down its modulus and argument.

**Exercise 3.6** (★★core — modulus and argument round-up). Suppose  $|z| = 2$ ,  $\text{Arg } z = \frac{\pi}{3}$ ,  $|w| = 3$ ,  $\text{Arg } w = \frac{\pi}{4}$ . Find the modulus and argument of (a)  $zw$ , (b)  $z/w$ , (c)  $z^2$  and (d)  $\bar{z}$ ; (e) convert  $z$  and  $z^2$  to Cartesian form.

**Exercise 3.7** (★★★challenge — the parallelogram law). Prove the parallelogram law: for all  $z, w \in \mathbb{C}$ ,

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2.$$

**Exercise 3.8** (★★★challenge — a square root in Cartesian form). Find all  $z = x + yi$  (with  $x, y \in \mathbb{R}$ ) such that  $z^2 = 3 + 4i$ , working directly in Cartesian form.